

DELAWARE COUNTY AP, IN, USA

WMO: 725336

Lat: 40.234N

Lon: 85.394W

Elev: 286

StdP: 97.94

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 94895

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.3	-14.1	-21.4	0.6	-16.5	-18.6	0.7	-13.5	11.7	-1.0	10.8	-1.1	3.4	270	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	32.4	23.1	31.2	22.7	29.9	22.0	25.1	30.1	24.2	28.8	23.5	28.0	4.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.8	28.2	22.6	18.0	27.3	22.1	17.4	26.5	77.6	30.0	74.3	29.1	71.3	27.9	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.0	8.1	-20.4	34.1	3.6	2.1	-22.9	35.6	-25.0	36.8	-27.0	37.9	-29.6	39.4		
(5)			-20.8	26.1	3.4	0.9	-23.2	26.8	-25.2	27.3	-27.2	27.9	-29.6	28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-2.5	-0.9	5.0	11.6	17.5	22.1	23.4	22.3	19.1	12.7	6.2	0.3		
	DBStd	10.56	6.91	6.64	6.36	5.41	4.92	3.52	3.12	2.97	4.18	5.30	5.51	6.09		
	HDD10.0	1408	390	311	183	44	3	0	0	0	1	31	139	306		
	HDD18.3	3057	646	539	416	210	77	9	2	4	41	191	363	559		
	CDD10.0	1941	2	5	28	93	237	364	417	381	272	113	26	4		
	CDD18.3	549	0	0	2	8	52	123	160	126	62	14	1	0		
	CDH23.3	4762	0	0	9	76	448	1031	1459	1041	582	114	2	0		
	CDH26.7	1474	0	0	1	8	124	336	496	296	186	27	0	0		
(14) Wind	WSAvg	3.8	4.7	4.6	4.5	4.6	3.8	3.3	2.8	2.5	2.7	3.6	4.2	4.4		
(15) Precipitation	PrecAvg	1061	67	59	75	102	121	120	113	92	81	75	84	72		
	PrecMax	1441	195	134	161	175	190	234	287	213	205	179	190	203		
	PrecMin	793	20	18	19	31	29	21	48	28	11	30	31	4		
	PrecStd	160	36	31	36	40	40	50	56	43	46	38	39	40		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.8	18.6	24.6	27.8	32.0	33.4	34.3	32.5	33.1	29.7	22.4	17.3		
		MCWB	13.2	12.3	16.6	18.1	20.9	22.6	25.0	23.7	22.4	20.9	15.7	13.8		
	2%	DB	12.8	15.3	20.8	25.8	29.6	32.0	32.3	31.2	31.1	26.4	19.2	14.0		
		MCWB	10.7	12.3	14.5	16.7	19.8	22.0	24.1	23.5	21.8	18.3	14.0	11.9		
	5%	DB	10.3	12.2	17.7	23.3	27.9	30.3	31.0	29.7	28.8	23.8	17.2	11.5		
		MCWB	8.3	9.6	12.6	15.5	19.2	21.5	23.5	22.6	20.9	17.3	13.2	9.4		
	10%	DB	7.2	8.8	15.1	21.1	26.0	28.6	29.4	28.2	26.9	21.7	14.9	8.8		
		MCWB	4.9	5.9	11.0	14.5	18.8	20.7	22.7	22.0	19.7	16.3	11.2	6.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.5	15.0	17.0	19.9	22.8	24.8	26.3	25.7	24.1	22.0	17.4	14.7		
		MCDB	15.4	17.5	23.7	25.9	28.3	30.5	31.9	30.5	30.0	27.8	20.2	16.6		
	2%	WB	11.1	12.4	15.3	18.2	21.5	23.8	25.4	24.7	23.1	20.5	15.4	12.0		
		MCDB	12.5	14.4	19.3	23.3	27.3	29.1	30.5	29.3	28.3	24.7	18.3	13.6		
	5%	WB	8.5	9.4	13.8	16.7	20.5	23.0	24.6	23.9	22.1	18.8	13.7	9.6		
		MCDB	10.1	12.1	17.2	21.4	26.0	28.3	29.4	28.0	27.0	22.3	16.4	11.5		
	10%	WB	5.0	5.9	11.5	15.3	19.7	22.1	23.8	23.0	21.0	16.8	11.6	6.9		
		MCDB	7.1	8.5	14.6	20.0	24.5	26.8	28.2	26.8	24.8	20.7	14.8	8.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	8.2	9.6	11.3	10.7	10.2	10.2	10.7	12.0	10.7	8.8	7.1		
		MCDBR	10.0	11.5	12.4	13.7	12.6	11.7	10.7	11.3	13.8	13.0	11.2	9.4		
		MCWBR	8.8	9.4	8.0	7.6	5.8	5.0	4.6	5.3	6.1	6.5	7.7	8.3		
		MCWBR	9.8	11.0	11.0	11.5	10.6	9.4	9.4	9.4	10.8	10.3	10.1	9.0		
(40) Clear-Sky Solar Irradiance	taub	taub	0.299	0.326	0.346	0.408	0.456	0.442	0.448	0.444	0.410	0.358	0.325	0.301		
		taud	2.416	2.324	2.359	2.192	2.100	2.195	2.183	2.182	2.252	2.396	2.441	2.444		
		Ebn at Noon	865	884	906	865	828	839	829	821	826	839	826	832		
		Edh at Noon	81	103	111	141	158	144	145	141	123	95	79	74		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.69	2.56	3.60	4.72	5.44	6.11	6.06	5.46	4.46	3.08	2.03	1.40		
		RadStd	0.17	0.28	0.29	0.43	0.48	0.42	0.35	0.29	0.37	0.28	0.24	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+189	+137
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+96	N/A

Nomenclature: See separate page